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Out-the-front knife

Abstract

An out-the-front knife, components thereof, and associated methods. The out-the front knife can be a double-action spring-assisted knife. The knife includes a handle and a blade. The handle includes a rear end and a front opening opposite the rear end. The blade is movable with respect to the handle between stowed and deployed positions. In the stowed position, the blade is biased rearward. The knife includes an actuator movable in a deploying direction to spring the blade forward away from the stowed position toward the deployed position.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of U.S. application Ser. No. 17/219,289, filed Mar. 31, 2021, which claims priority to U.S. Provisional Application Nos. 63/002,788, filed Mar. 31, 2020, and 63/034,597, filed Jun. 4, 2020, the entireties of which are hereby incorporated by reference.

FIELD

(1) The present disclosure generally relates to knives, and more particularly to out-the-front knives or knives having a handle and a blade that is deployable out a front end of the handle.

BACKGROUND

(2) People use deployable blade knives in many situations. Such knives have handles configured to

receive a blade of the knife in a stowed position. The blade can be moved from the stowed position to a deployed position when the user desires to use the knife. In some cases, the blade is spring-biased to move the blade to the deployed position for ease of use.

SUMMARY

(3) In one aspect, an out-the-front knife comprises a handle having a rear end and a front opening opposite the rear end. A blade supported by the handle has a cutting edge. The blade is movable with respect to the handle between a stowed position in the handle and a deployed position in which the blade extends out of the front opening. An actuator supported by the handle is movable with respect to the handle in a deploying direction to move the blade forward away from the stowed position toward the deployed position. The actuator being movable in a stowing direction to move the blade rearward away from the deployed position toward the stowed position. A blade driver supported by the handle is configured to spring the blade toward the deployed position responsive to movement of the actuator in the deploying direction. The blade driver is configured to spring the blade toward the stowed position responsive to movement of the actuator in the stowing direction. The blade when in the stowed position is biased rearward. The blade driver includes a blade pusher configured to push the blade forward against said rearward bias responsive to movement of the actuator in the deploying direction. The blade driver being configured to spring the blade toward the deployed position after sufficient forward pushing force is applied to the blade by the blade pusher to overcome the rearward bias.

(4) In another aspect, an out-the-front knife comprises a handle having a rear end and a front opening opposite the rear end. A blade supported by the handle has a cutting edge. The blade is movable with respect to the handle between a stowed position in the handle and a deployed position in which the blade extends out of the front opening. The blade when in the stowed position is biased rearward. An actuator supported by the handle is movable with respect to the handle in a deploying direction to move the blade forward away from the stowed position toward the deployed position. A blade driver supported by the handle includes at least one spring for springing the blade away from the stowed position toward the deployed position. The blade driver includes a first blade pusher and a second blade pusher different from the first blade pusher. The first blade pusher is operatively connected to the spring to transfer spring force from the spring to the blade to spring the blade toward the deployed position. The second blade pusher is configured to push the blade forward against the rearward bias responsive to movement of the actuator in the deploying direction. The first pusher is configured to spring the blade toward the deployed position via spring force from the spring after sufficient forward pushing force is applied to the blade by the second blade pusher to overcome the rearward bias.

(5) In yet another aspect, an out-the-front knife comprises a handle having a rear end and a front opening opposite the rear end. A blade supported by the handle has a cutting edge. The blade is movable with respect to the handle between a stowed position in the handle and a deployed position in which the blade extends out of the front opening. A retainer is configured to engage the blade in rearward biasing engagement to bias the blade rearward when the blade is in the stowed position. An actuator supported by the handle is movable with respect to the handle in a deploying direction. The actuator is operatively connected to the blade to cause the blade to move forward away from the stowed position toward the deployed position responsive to movement of the actuator in the deploying direction. The blade includes a ramp configured to move the retainer out of rearward biasing engagement with the blade when sufficient force is applied to the actuator in the deploying direction to overcome the rearward bias.

(6) Other objects and features of the present disclosure will be in part apparent and in part pointed out herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a front top perspective of a knife of the present disclosure showing a blade of the knife in a deployed position;
- (2) FIG. 2 is a perspective of the knife similar to FIG. 1 but showing the blade in a stowed position;
- (3) FIG. 3 is an exploded perspective of the knife;
- (4) FIG. 4 is a front top perspective of the knife having a top outer handle portion removed to expose interior components;
- (5) FIG. 5 is a top perspective of a slider of the knife;
- (6) FIG. 6 is a bottom perspective of the slider;
- (7) FIG. 7 is a top perspective of the slider with two blade movers and springs connecting the blade movers;
- (8) FIG. 8 is a top front perspective of another blade mover of the knife;
- (9) FIG. 9 is a perspective of a first retainer of the knife;
- (10) FIG. 10 is a front elevation of the knife having the top outer handle portion removed, with the blade shown in a stowed position;
- (11) FIG. 11 is an enlarged fragmentary section of the knife in the configuration of FIG. 10 showing the first retainer engaging the blade in rearward biasing engagement;
- (12) FIG. 12 is front elevation of the knife similar to FIG. 10 but showing an actuator and slider thereof moved forward;
- (13) FIG. 13 is a front elevation similar to FIG. 12 but showing the actuator and slider moved farther forward, and the first retainer just having been moved out of rearward biasing engagement with the blade;
- (14) FIG. 14 is a front elevation similar to FIG. 13 but showing the blade having moved to a deployed position, with a second retainer maintaining the blade in the deployed position;
- (15) FIG. 15 is a front elevation similar to FIG. 14 but showing the actuator and slider moved rearward, and a ramp of the slider almost releasing the second retainer to permit the blade to spring rearward to the stowed position; and
- (16) FIG. 16 is a rear perspective of the knife showing features of a clip of the knife.
- (17) Corresponding reference numbers indicate corresponding parts throughout the drawings.

DETAILED DESCRIPTION

- (18) Referring to FIG. 1, an out-the-front knife of the present disclosure is indicated by the reference number 10. In general, the knife includes a handle 12, a blade 14, and an actuator 16. The actuator 16 can be manipulated by a user to expose the blade 14 and hide the blade in the handle 12.
- (19) Referring to FIGS. 1 and 2, the handle 12 has a front end (facing to the left and out of the page), a rear end (facing to the right and into the page), and opposite major side faces and opposite minor side faces extending between the front and rear ends. The handle 12 has a front opening 12A at the front end through which the blade 14 is deployable. Thus, the knife 10 may be referred to as an “out-the-front” knife, meaning the blade 14 is deployable out of the front of the handle 12. As shown in FIGS. 2 and 3, the handle 12 of the illustrated knife 10 includes three handle portions 12B, 12C, 12D. Upper and lower 12B, 12C outer handle portions sandwich an intermediate handle portion 12D. The handle portions can be secured to each other using fasteners such as screws.
- (20) The handle 12 defines an interior space comprising a first cavity between the upper outer handle portion 12B and the intermediate handle portion 12D, for housing a first group of internal components (primarily components for driving movement of the blade 14), and a second cavity between the lower outer handle portion 12C and intermediate handle portion, for housing additional internal components (primarily the blade).
- (21) The blade 14 has at least one cutting edge 14A. The blade has a tip at its forward end, and includes a tang 14B at its rear end. The blade 14 includes two bosses 14C (broadly, “followers”)

protruding from opposite sides of the tang **14B**, and includes a notch **14D** (broadly, “recess”), for reasons which will be explained below. The bosses **14C** can be formed by a single cylindrical piece press fitted into a hole in the tang **14B**.

(22) The blade **14** is movable with respect to the handle **12** between a stowed position (e.g., FIG. 2) and a deployed position (e.g., FIG. 1). In the stowed position, the blade **14** is located in the handle **12**. The cutting edge **14A** is inside the handle **12** to protect the cutting edge and/or for safety while storing the knife. In the deployed position, the blade **14** extends out of the front opening **12A** of the handle **12** and the cutting edge **14A** is exposed for use.

(23) The bosses **14C** protruding from the tang **14B** are received in respective elongate channels **20** (broadly, “guides”), one of which is provided in the intermediate handle portion **12D**, and the other of which is provided in the lower outer handle portion **12C**. The elongate channels **20** serve as tracks to guide movement of the blade **14** with respect to the handle **12** and to locate the blade with respect to the handle in the stowed and deployed positions. Opposite sides of the blade tang **14B** may also engage opposite interior walls of the lower outer handle portion **12C** to guide movement of the blade **14** and maintain a longitudinal axis of the blade extending generally parallel with a longitudinal axis of the handle **12**.

(24) The actuator **16** is movable forward (broadly, in a deploying direction) with respect to the handle **12** to move the blade **14** forward away from the stowed position toward the deployed position. The actuator **16** is also movable rearward (broadly, in a stowing direction) to move the blade **14** rearward away from the deployed position toward the stowed position. These operations would normally be performed by one hand of a user while that hand holds the handle **12**. For example, while the user's hand holds the handle **12**, the thumb of that hand can be used to slide the actuator **16** forward or rearward to extend or retract the blade **14**.

(25) The knife **10** includes a blade driver **26** (broadly, “linkage”) linking the actuator **16** to the blade **14** to permit movement of the blade by the actuator. The blade driver **26** is configured to spring the blade **14** toward the deployed position responsive to movement of the actuator **16** in the deploying direction. The blade driver **26** is also configured to spring the blade **14** toward the stowed position responsive to movement of the actuator **16** in the stowing direction. Accordingly, the knife **10** can be referred to as a dual-action spring-assisted knife. Other configurations (e.g., single-action spring assisted) can be used without departing from the scope of the present disclosure.

(26) The blade driver **26** includes a slider **30**, two springs **32**, and three blade pushers **34A**, **34B**, **36**. The slider **30** is captured between the upper outer handle portion **12B** and the intermediate handle portion **12D**. The slider **30** is movable forward and rearward with respect to the handle **12** conjointly with the actuator **16**. The slider **30** includes a tab **30A** for connecting the actuator to the slider **16** (e.g., using roll pins **38**).

(27) Referring to FIG. 5, the slider **30** defines an elongate channel **30B** serving as a bed for the two springs **32**. Opposite ends of the springs **32** are connected to the first blade pusher **34A** and the third blade pusher **34B**. The first and third blade pushers **34A**, **34B** have holes therein for receiving hooked ends of the springs **32**. Although some of the drawings may not show the springs **32** connected to both of the first and third blade pushers **34A**, **34B**, it will be appreciated that such connections would be made in manufacture. For example, the extension springs **32** may be sized such that when the blade driver **26** is at rest the springs are in some tension, forcing the first and third blade pushers **34A**, **34B** against the opposite ends of the slider **30**.

(28) The illustrated first and third blade pushers **34A**, **34B** comprise brackets including flanges **34A'**, **34B'** extending downward below the slider for engagement with the blade **14**. The first and third pushers **34A**, **34B** transmit spring force from the two springs **32** to the blade **14** to spring the blade toward the deployed position and to spring the blade toward the stowed position. In particular, the first pusher **34A** pushes against the rear end of the tang **14B** to deploy the blade **14**, and the third pusher **34B** pushes against a forward end of the boss **14C** to retract the blade.

(29) The second blade pusher **36** has a different construction and function from the first and third blade pushers **34A**, **34B**. The second blade pusher **36** is not connected to the springs **32**. The second blade pusher **36** includes a forward head **36A** and a rear foot **36B**. The head **36A** is arranged to be pushed forward by a tab **30C** of the slider **30** when the slider is pushed forward. The foot **36B** is arranged to be located behind the blade tang **14B** for pushing the blade forward when the slider **30** pushes the second blade pusher **36** forward. Operation and function of the second blade pusher **36** will be explained in more detail below. In the illustrated embodiment, the second blade pusher **36** includes a bar **36C** extending between the head **36A** and the foot **36B**, which are formed by flanges bent with respect to the bar. Other configurations can be used without departing from the scope of the present disclosure.

(30) The knife **10** includes a first retainer **40** configured to retain the blade in the stowed position. In the illustrated embodiment, the retainer **40** comprises a spring **42** and a detent **44** biased by the spring. The detent **44** includes a cam **44A** arranged to cam against the notch **14D** in the blade tang. The cam **44A** includes a rounded cam surface that is received in the notch **14D** when the blade **14** is in the stowed position. The blade tang **14B** includes a ramp **14D'** partially defining the notch, and the cam engages the ramp in the stowed position. As shown in FIG. **11**, when the blade **14** is in the stowed position, the cam **44A** does not “bottom out” in the notch **14D**, and thus remains in engagement with the ramp **14D'**. The engagement of the cam **44A** with the ramp **14D'** can be referred to broadly as rearward biasing engagement. The lateral spring bias of the cam **44A** is converted to rearward bias on the blade **14** by the engagement of the cam with the ramp **14D'**. There is not sufficient clearance in the rear end of the handle for the tang **14B** to move sufficiently rearward for the cam **44A** to bottom out in the notch **14D**, thus the cam remains in contact with the ramp **14D'** when the blade **14** is in the retracted position, biasing the blade toward closure. If the actuator **16** were pushed partially forward and then released before overcoming the rearward bias, the cam **44A** would ride laterally outward on the ramp **14D'** and then force the blade **14** rearward back to the stowed position when the actuator was released.

(31) The knife **10** includes a second retainer **50** configured to retain the blade **14** in the deployed position. In the illustrated embodiment, the second retainer **50** comprises a lever **50A** biased by a spring **50B** to an extended position in which the lever extends at a skew angle with respect to the longitudinal axis of the handle **12**. The lever **50A** is located to engage the blade tang **14B** to “lock” the blade in the deployed position. The arrangement is such that as the blade **14** moves forward, the front of the tang **14B** engages the lever **50A** and causes it to temporarily deflect against the spring bias. When the tang **14B** passes the lever **50A**, it resiliently extends to its extended position, in which it prevents the tang from retracting into the handle **12**. As shown in FIG. **6**, the slider includes a second ramp **30D** arranged to, as the actuator **16** is moved rearward, engage the lever **50A** to move the lever out of retaining engagement with the rear end of the tang **14B**, to permit the blade **14** to move toward the stowed position.

(32) A cycle of use of the knife **10** will now be discussed with respect to FIGS. **10-15**. In a first stage, shown in FIG. **10**, the blade **14** is in the stowed position in the handle **12**. When the user desires to deploy the blade **14**, the user pushes the actuator **16** in the deploying direction (forward). As the actuator **16** travels a first distance **D1** in the deploying direction, the slide **30** moves forward, which moves the third blade pusher **34B** forward while the first blade pusher **34A** remains behind the blade tang **14B**, thus creating additional tension in the springs **32**. During movement of the actuator **16** in this first distance **D1**, the slider tab **30C** moves forward conjointly with the actuator **16**, but does not yet contact the second pusher head **36A** because there is a gap between the head and the tab. Referring to FIG. **12**, after the actuator travels the first distance **D1** forward, the tab **30C** contacts the head **36A** of the second blade pusher **36**, which begins pushing the blade **14** forward via the foot **36B** behind the tang **14B**. While the actuator **16** is moved the second distance in the deploying direction, the tension continues to build in the springs **32**, and the ramp **14D'** of the blade **14** starts to push the detent **44** to the side. The detent **44** rides laterally outward

on the ramp **14D'** as the blade **14** moves forward, but still biases the blade toward the stowed position (toward closure). If the actuator **16** were released by the user before the actuator travels the full second distance **D2**, the detent **44** would resiliently spring back, pushing against the ramp **14D** and causing the blade **14** to resume the stowed position. But if the actuator **16** is pushed the full second distance **D2**, that means sufficient force has been applied by the user to the actuator to overcome the rearward bias, and the ramp **14D'** has pushed the detent **44** out of rearward biasing engagement with the blade **14**. The blade **14** is thus free to move forward and is flung forward by the spring force of the springs **32** transmitted to the blade **14** via the first blade pusher **34A**. The tang **14B** temporarily deflects the second retainer **50**, until the tang passes the retainer, which then resiliently extends to lock the blade **14** in the deployed position. It will be appreciated that the knife **10** is free of structure, other than the blade **14** (specifically, the ramp **14D'**), configured to engage the first retainer **40** to move the retainer out of rearward biasing engagement with the blade. For example, the slider **30** does not include structure that engages the first retainer **40** to move it out of rearward biasing engagement with the blade **14**.

(33) When the user desires to retract the blade **14** to the stowed position, the user applies manual force to the actuator **16** in the stowing direction (rearward). In a first stage of rearward travel of the actuator **16**, tension builds in the springs **32**. In a second stage of rearward travel of the actuator **16**, tension continues to build in the springs **32**, and the ramp **30D** of the slider **30** contacts the lever **50A** of the second retainer **50** and begins moving it out of retaining engagement with the tang **14B**. Eventually, the second ramp **30D** moves the retainer **50** against its spring bias fully out of retaining engagement with the blade **14**, permitting the third blade pusher **34B** to fling the blade toward its stowed position via the spring force from the springs **32**. As the blade **14** moves rearward, the tang **14B** deflects the first retainer **50** until the tang moves sufficiently rearward to permit the first retainer to resiliently extend into the notch **14D**, causing the retainer to once again engage the ramp **14D'** to maintain the blade **14** in the stowed position and bias the blade rearward while in the stowed position.

(34) The knife **10** includes a clip **60** configured to facilitate connecting the knife to a pocket or a belt of a user. The clip **60** includes an arm **62** secured to the handle **12** by two fasteners **64** threaded into threaded openings in the rear end of the lower outer handle portion **12B**. The arm **62** includes a mounting portion **62A** and a main body **62B** extending generally transversely with respect to the main mounting portion. The mounting portion **62A** has two slots **62C** in which the fasteners **64** are received. The fasteners **64** are tightened to hold the arm **62** in position with respect to the handle **12**, with the mounting portion **62A** pressed against the rear end of the handle. The slots **62C** permit the arm **62** to be adjusted (inward or outward relative to the handle **12**) to apply more or less pressing force of the arm against the side of the handle per user preference. When the arm **62** is in the desired configuration, the fasteners **64** are tightened to hold that configuration. Moreover, the slots **62C** permit the clip **60** to be reversible such that the arm **62** can be on the top or bottom of the handle **12**. It will be appreciated that the openings receiving the fasteners **64** are offset to the bottom of the handle **12**. Accordingly, if the arm **62** is to be mounted to the top side of the handle **12**, the mounting portion **62A** will need to reach farther to the fastener openings, and the slots **62C** permit passage of the fasteners **64** through the mounting portion **62A** in such a configuration. The rear ends of the upper and lower handle portions **12A**, **12B** and of the intermediate handle portion **12C** have notches therein to permit generally flush or recessed seating of the mounting portion **62A** of the arm **62** against the rear end of the handle **12**. Engagement of sides of the mounting portion **62A** with side walls of the notches limits rotation of the arm **62** and helps maintain the position of the arm on the handle **12**.

(35) It will be apparent that modifications and variations are possible without departing from the scope of the invention defined in the appended claims.

(36) As various changes could be made in the above constructions and methods without departing

from the scope of the invention, it is intended that all matter contained in the above description and shown in the accompanying drawings shall be interpreted as illustrative and not in a limiting sense.

Claims

1. An out-the-front knife comprising: a handle having a rear end and a front opening opposite the rear end; a blade supported by the handle, the blade having a cutting edge, the blade being movable with respect to the handle between a stowed position in the handle and a deployed position in which the blade extends out of the front opening; and an actuator supported by the handle, the actuator being movable with respect to the handle in a deploying direction to move the blade forward away from the stowed position toward the deployed position; a blade driver supported by the handle, the blade driver being configured to spring the blade toward the deployed position responsive to movement of the actuator in the deploying direction; wherein the blade when in the stowed position is biased rearward, and the blade driver includes a blade pusher configured to push the blade forward against said rearward bias responsive to movement of the actuator in the deploying direction, the blade driver being configured to spring the blade toward the deployed position after sufficient forward pushing force is applied to the blade by the blade pusher to overcome the rearward bias.
2. An out-the-front knife as set forth in claim 1, further comprising a retainer configured to engage the blade in rearward biasing engagement to bias the blade rearward in the stowed position.
3. An out-the-front knife as set forth in claim 2, wherein the blade is configured to push the retainer out of rearward biasing engagement with the blade when sufficient force is applied to the actuator in the deploying direction.
4. An out-the-front knife as set forth in claim 3, wherein the blade includes a ramp configured to push the retainer out of rearward biasing engagement with the blade when sufficient force is applied to the actuator in the deploying direction.
5. An out-the-front knife as set forth in claim 3, wherein the knife is free of structure, other than the blade, configured to engage the retainer to move the retainer out of rearward biasing engagement with the blade.
6. An out-the-front knife as set forth in claim 2, wherein the retainer comprises a cam configured to cam against the blade to convert a lateral bias of the cam to the rearward bias of the blade in the stowed position.
7. An out-the-front knife as set forth in claim 6, wherein the blade includes a ramp and the cam is configured to engage the ramp to bias the blade rearward when the blade is in the stowed position.
8. An out-the-front knife as set forth in claim 2, wherein the retainer is retractable from a retaining position in engagement with the blade to a retracted position, the retainer being configured for non-pivoting movement from the retaining position to the retracted position.
9. An out-the-front knife as set forth in claim 2, wherein the retainer is retractable from a retaining position in engagement with the blade to a retracted position, the retainer being configured for generally linear movement from the retaining position to the retracted position.
10. An out-the-front knife as set forth in claim 1, wherein the blade driver includes at least one spring, and the blade driver is configured to create spring force in the spring to spring the blade toward the deployed position responsive to movement of the actuator in the deploying direction.
11. An out-the-front knife as set forth in claim 10, wherein the blade driver is configured to, responsive to movement of the actuator a first distance in the deploying direction, create spring force in the at least one spring, and the blade driver is configured to, responsive to movement of the actuator beyond the first distance in the deploying direction, cause the blade pusher to begin pushing the blade forward toward the deployed position.
12. An out-the-front knife as set forth in claim 1, wherein the blade pusher is a first blade pusher, the blade driver including a spring for springing the blade away from the stowed position toward

the deployed position, the blade driver including a second blade pusher different from the first blade pusher, the second blade pusher being operatively connected to the spring to transfer spring force from the spring to the blade to spring the blade toward the deployed position, the second pusher being configured to spring the blade toward the deployed position via spring force from the spring after sufficient forward pushing force is applied to the blade by the first blade pusher to overcome the rearward bias.

13. An out-the-front knife as set forth in claim 12, wherein the blade driver is configured to, responsive to movement of the actuator a first distance in the deploying direction, create spring force in the spring for springing the blade toward the deployed position, and the blade driver is configured to, responsive to movement of the actuator beyond the first distance in the deploying direction, cause the first blade pusher to begin pushing the blade forward toward the deployed position.

14. An out-the-front knife as set forth in claim 12, wherein the actuator is movable with respect to the handle in a stowing direction to move the blade rearward away from the deployed position toward the stowed position.

15. An out-the-front knife as set forth in claim 12, further comprising a retainer configured to engage the blade in rearward biasing engagement to bias the blade rearward in the stowed position.

16. An out-the-front knife as set forth in claim 15, wherein the blade is configured to push the retainer out of rearward biasing engagement with the blade when sufficient force is applied to the actuator in the deploying direction.

17. An out-the-front knife as set forth in claim 16, wherein the blade includes a ramp configured to push the retainer out of rearward biasing engagement with the blade when sufficient force is applied to the actuator in the deploying direction to overcome the rearward bias.

18. An out-the-front knife as set forth in claim 16, wherein the knife is free of structure, other than the blade, configured to engage the retainer move the retainer out of rearward biasing engagement with the blade.

19. An out-the-front knife as set forth in claim 15, wherein the retainer comprises a cam configured to cam against the blade to convert a lateral bias of the cam to the rearward bias of the blade in the stowed position.

20. An out-the-front knife as set forth in claim 1, wherein when the blade is in the stowed position, the actuator is in an at-rest position from which the actuator is movable in the deploying direction.
